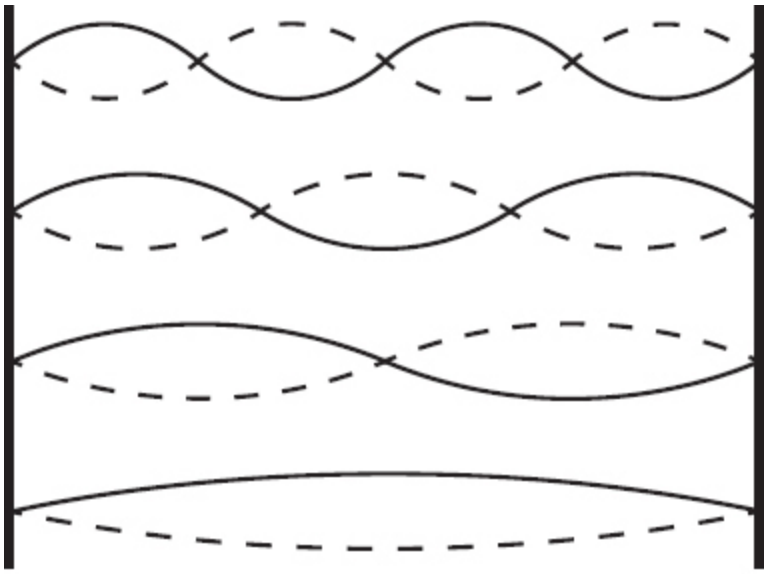


Allowed wavelengths (modes)
of a string with ends tied down

The word “quantum,” referring to some definite amount of stuff, can give the impression that quantum mechanics describes a world that is fundamentally discrete and pixelated, like when you zoom in closely on a computer monitor or TV screen. It’s actually the opposite; quantum mechanics describes the world as a smooth wave function. But in the right circumstances, where individual parts of the wave function are tied down in a certain way, the wave takes the form of a combination of distinct vibrational modes. When we observe such a system, we see those discrete possibilities. That’s true for orbits of electrons, and it will also explain why quantum fields look like sets of individual particles. In quantum mechanics, the world is fundamentally wavy; its apparent quantum discreteness comes from the particular way those waves are able to vibrate.

De Broglie’s ideas were intriguing, but they fell short of providing a comprehensive theory. That was left to Erwin Schrödinger, who in 1926 put forth a dynamical understanding of wave functions, including the equation they obey, later named after him. Revolutions in physics are generally a young person’s game, and quantum mechanics was no different, but Schrödinger bucked the trend. Among the leaders of the discussions at Solvay in 1927, Einstein at forty-eight years old, Bohr at forty-two, and Born at forty-four were the grand old men. Heisenberg was twenty-five, Pauli twenty-seven, and Dirac twenty-five. Schrödinger, at the ripe old age